

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403898
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	McGrory; Jackson Barry et al.

Method and system for friction-aware impact avoidance

Abstract

A method controlling a vehicle includes detecting that an object is along a path of the vehicle and within a predetermined distance from the vehicle, in response to detecting the object that is along the path of the vehicle and within the predetermined distance from the vehicle, determining, in real time, a coefficient of friction between a road and a tire of the vehicle, and in response to determining the coefficient of friction between the road and the tire of the vehicle, commanding the vehicle to perform a control action.

Inventors: McGrory; Jackson Barry (Mississauga, CA), Otanez; Paul Guillermo (Franklin, MI), Manogaran; Ravichandran (Ann Arbor, MI), Shahriari; Mohammadali (Markham, CA)

Applicant: GM Global Technology Operations LLC (Detroit, MI)

Family ID: 94875385

Assignee: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Appl. No.: 18/475621

Filed: September 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250100547 A1	Mar. 27, 2025

Publication Classification

Int. Cl.: B60W30/09 (20120101); B60W10/18 (20120101); B60W10/20 (20060101); B60W30/095 (20120101); B60W40/068 (20120101); B60W50/12 (20120101)

U.S. Cl.:

CPC **B60W30/09** (20130101); **B60W10/18** (20130101); **B60W10/20** (20130101);
B60W30/0956 (20130101); **B60W40/068** (20130101); **B60W50/12** (20130101);
B60W2552/40 (20200201); B60W2554/80 (20200201)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8788176	12/2013	Yopp	701/96	B60W 30/162
11097739	12/2020	Capua	N/A	B60W 40/107
12195082	12/2024	Schumann	N/A	B62D 15/025
2007/0061061	12/2006	Salman	701/72	B60T 8/172
2008/0046145	12/2007	Weaver	701/41	B60T 7/22
2012/0022747	12/2011	Zagorski	701/80	B60T 8/172
2013/0211687	12/2012	Trost	701/70	B60T 7/22
2016/0167519	12/2015	Luke	701/22	B60L 58/12
2018/0037234	12/2017	Hoedt	N/A	G05D 1/0891
2019/0126933	12/2018	Jonasson	N/A	B60W 40/072
2019/0256103	12/2018	Capua	N/A	B60W 40/101
2021/0383040	12/2020	Cho	N/A	B60T 17/22
2022/0185293	12/2021	Choi	N/A	B60W 10/18
2022/0212677	12/2021	Zheng	N/A	B60T 8/17551
2023/0211788	12/2022	Zhang	701/80	B60T 8/172

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112884288	12/2020	CN	G06Q 50/265

OTHER PUBLICATIONS

Translation and Original for CN112884288A, Jun. 1, 2021. cited by examiner

Primary Examiner: Bendidi; Rachid

Assistant Examiner: Alqaderi; Nada Mahyoob

Attorney, Agent or Firm: Vivacqua Crane, PLLC

Background/Summary

INTRODUCTION

(1) The present disclosure relates to a method and system for controlling a vehicle. More specifically, the present disclosure relates to a method and system for friction-aware impact avoidance.

(2) This introduction generally presents the context of the disclosure. Work of the presently named

inventors, to the extent it is described in this introduction, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against this disclosure.

(3) The motion of a vehicle may be affected by the friction between the road and the vehicle tires. Therefore, it is useful to know the coefficient of friction between the road surface and the vehicle tires. In addition, it is desirable to consider the friction between the road and the vehicle tires when controlling the movement of the vehicle.

SUMMARY

(4) The present disclosure describes a method for friction-aware impact avoidance. In an aspect of the present disclosure, the method includes detecting that an object is along a path of the vehicle and within a predetermined distance from the vehicle, determining, in real time, a coefficient of friction between a road and a tire of the vehicle in response to detecting the object that is along the path of the vehicle and within the predetermined distance from the vehicle, and commanding the vehicle to perform a control action in response to determining the coefficient of friction between the road and the tire of the vehicle. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods. The method described in this paragraph improves vehicle technology by using the coefficient of friction between the road and the tires of the vehicles to avoid impacting objects.

(5) Implementations may include one or more of the following features. The method may include applying a braking load to a wheel of the vehicle, monitoring a movement of the vehicle (e.g., longitudinal deceleration, wheel speed and/or GNSS speed) at the same time as the braking load is applied to the wheel of the vehicle, and determining, in real time, the coefficient of friction between the road and the tire of the vehicle based on the movement of the vehicle while the braking load. The control action may be continuously applying the braking load to the wheel of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is less than the predetermined friction threshold. Alternatively, or additionally, the control action may be steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold. The steering path has a radius of curvature. The method may include a radius of curvature of the steering path. The radius of curvature of the steering path may be based on the coefficient of friction between the tire and the road. The method may include determining a time to impact with the object that is along the path of the vehicle and within the predetermined distance from the vehicle, determining a last time to determine friction; determining a last time to steer the vehicle to avoid the object along the path of the vehicle, determining a sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle, and comparing the time to impact with the object that is along the path of the vehicle with the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle to determine whether the time to impact with the object that is along the path of the vehicle is greater than the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle. The method may include determining the last time to brake to avoid the object along the path of the vehicle in response to determining that the time to impact with the object that is along the path of the vehicle is greater than the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle. The method may include determining a sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction in response to determining that the time to impact with the object that is along the path of the vehicle is greater than the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle. Further, the method may include comparing the time to impact with the object that is along the path of the vehicle with

the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction to determine whether the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction in response to determining that the time to impact with the object that is along the path of the vehicle is greater than the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle. The coefficient of friction between the road and the tire of the vehicle may be determined solely in response to: (a) detecting the object that is along the path of the vehicle and within the predetermined distance from the vehicle; and (b) determining that the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction. The method may include determining an updated last time to steer to avoid the object that is along the path of the vehicle based on the coefficient of friction between the road and the tire of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold and comparing the time to impact with the object along the path of the vehicle with the updated last time to steer to avoid the object that is along the path of the vehicle to determine whether the time to impact with the object along the path of the vehicle is greater than the updated last time to steer to avoid the object that is along the path of the vehicle. The vehicle may be commanded to perform the control action solely in response to determining that the time to impact with the object along the path of the vehicle is greater than the updated last time to steer to avoid the object that is along the path of the vehicle. The control action may be steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold. The control action may include commanding a display of the vehicle to show an alert, such as “keep braking” and/or “do not steer”). The display **29**, for example, may issue an alert to the driver (e.g., “keep braking” and/or “do not steer”). Inadequate road friction may be used as a condition to inhibit prompts or “nudges” to suggest that the driver attempt to steer evasive (e.g., alarms, vibrations, screen messages, HUD alerts, etc.). Inadequate road friction may be used as a condition to inhibit evasive steering assist that engages after the driver begins to steer.

Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(6) The present disclosure also describes a system including sensors and a controller in communication with the sensors. The controller is programmed to execute the method described above. The present disclosure also describes a tangible, non-transitory, machine-readable medium, comprising machine-readable instructions, that when executed by a processor, cause the processor to execute the method described above.

(7) Further areas of applicability of the present disclosure will become apparent from the detailed description provided below. It should be understood that the detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

(8) The above features and advantages, and other features and advantages, of the presently disclosed system and method are readily apparent from the detailed description, including the claims, and exemplary embodiments when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. **1** is a block diagram of a vehicle.

(3) FIG. 2 is a flowchart of a method for friction-aware impact avoidance.

(4) FIG. 3 is a schematic diagram of the vehicle approaching an object.

DETAILED DESCRIPTION

(5) Reference will now be made in detail to several examples of the disclosure that are illustrated in accompanying drawings. Whenever possible, the same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps.

(6) With reference to FIG. 1, a vehicle **10** is configured to be driven on a road **39** and generally includes a chassis **12**, a body **14**, wheels **15**, front and rear tires **17** and may be referred to as a vehicle system. In the depicted embodiment, the vehicle **10** includes two front tires **17a** and two rear tires **17b**. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The wheels **15** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**. Each tire **17** is coupled to one of the wheels **15**. The vehicle **10** includes a front axle **19** coupled to the front tires **17a** and a rear axle **25** coupled to the rear tires **17b**.

(7) The vehicle **10** may be an autonomous vehicle, and a control system **98** is incorporated into the vehicle **10**. The vehicle **10** is, for example, a vehicle that is automatically controlled to carry passengers from one location to another. The vehicle **10** is depicted in the illustrated embodiment as a pickup truck, but it should be appreciated that other vehicles including, trucks, sedans, coupes, sport utility vehicles (SUVs), recreational vehicles (RVs), etc., may also be used. In an embodiment, the vehicle **10** may include a so-called a Level Two, a Level Three, Level Four, or Level Five driving automation system. A Level Four system indicates “high automation,” referring to the driving mode-specific performance by an automated driving system of aspects of the dynamic driving task, even if a human driver does not respond appropriately to a request to intervene. A Level Five system indicates “full automation,” referring to the full-time performance by an automated driving system of aspects of the dynamic driving task under a number of roadway and environmental conditions that can be managed by a human driver. In Level 3 vehicles, the system **98** performs the entire dynamic driving task (DDT) within the area that it is designed to do so. In Level 2 vehicles, systems provide steering, brake/acceleration support, lane centering, and adaptive cruise control. However, even if these systems are activated, the vehicle operator at the wheel must be driving and constantly supervising the automated features.

(8) As shown, the vehicle **10** generally includes a propulsion system **20**, a transmission system **22**, a steering system **24**, a brake system **26**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, at least one controller **34**, and a communication system **36**. The steering system **24** is a steer-by-wire system or electronic power steering. The propulsion system **20** may, in various embodiments, include an electric machine such as a traction motor and/or a fuel cell propulsion system. The vehicle **10** may further include a battery (or battery pack) **21** electrically connected to the propulsion system **20**. Accordingly, the battery **21** is configured to store electrical energy and to provide electrical energy to the propulsion system **20**. In certain embodiments, the propulsion system **20** may include an internal combustion engine. The transmission system **22** is configured to transmit power from the propulsion system **20** to the vehicle tires **17** according to selectable speed ratios. According to various embodiments, the transmission system **22** may include a step-ratio automatic transmission, a continuously-variable transmission, or other appropriate transmission. The brake system **26** is configured to provide braking torque to the vehicle tires **17**. The brake system **26** may, in various embodiments, include friction brakes, brake by wire, a regenerative braking system such as an electric machine, and/or other appropriate braking systems. The steering system **24** influences the position of the vehicle tires **17** and may include a steering wheel **33**. While depicted as including a steering wheel **33** for illustrative purposes, in some embodiments contemplated within the scope of the present disclosure, the steering system **24** may not include a steering wheel **33**.

(9) The sensor system **28** includes one or more sensors **40** (i.e., sensing devices) that sense

observable conditions of the exterior environment and/or the interior environment of the vehicle **10**. The sensors **40** are in communication with the controller **34** and may include, but are not limited to, one or more steering wheel sensors **45**, one or more radars, one or more light detection and ranging (lidar) sensors, one or more proximity sensors, one or more odometers, one or more ground penetrating radar (GPR) sensors, one or more steering angle sensors, Global Navigation Satellite System (GNSS) transceivers (e.g., one or more global positioning systems (GPS) transceivers), one or more tire pressure sensors, one or more cameras **41** (e.g., eye tracker), one or more gyroscopes, one or more accelerometers, one or more inclinometers, one or more speed sensors, one or more ultrasonic sensors, one or more inertial measurement units (IMUs), one or more night-vision devices, thermal imaging sensors, and/or other sensors. Each sensor **40** is configured to generate a signal that is indicative of the sensed observable conditions of the exterior environment and/or the interior environment of the vehicle **10**. Because the sensor system **28** provides data to the controller **34**, the sensor system **28** and its sensors **40** are considered sources of information (or simply sources). The vehicle **10** and/or system **98** does not include light sensors capable of detecting light inside the vehicle **10**.

(10) The actuator system **30** includes one or more actuator **42** that control one or more vehicle features such as, but not limited to, the propulsion system **20**, the transmission system **22**, the steering system **24**, and the brake system **26**. The steering system **24** may include a front and possibly rear steer. The rear steering control may be controlled by a separate controller. In various embodiments, the vehicle features may further include interior and/or exterior vehicle features such as, but are not limited to, doors, a trunk, and cabin features such as air, music, lighting, etc. The actuators **42** may be part of the steering system **24** and include one or more road wheel actuators (RWAs) and a hand wheel actuator (HWA).

(11) The data storage device **32** stores data for use in automatically controlling the vehicle **10**. In various embodiments, the data storage device **32** stores defined maps of the navigable environment. In various embodiments, the defined maps may be predefined by and obtained from a remote system. For example, the defined maps may be assembled by the remote system and communicated to the vehicle **10** (wirelessly and/or in a wired manner) and stored in the data storage device **32**. The data storage device **32** may be part of the controller **34**, separate from the controller **34**, or part of the controller **34** and part of a separate system.

(12) The vehicle **10** may further include one or more airbags **35** in communication with the controller **34** or another controller of the vehicle **10**. The airbag **35** includes an inflatable bladder and is configured to transition between a stowed configuration and a deployed configuration to cushion the effects of an external force applied to the vehicle **10**. The sensors **40** may include an airbag sensor, such as an IMU, configured to detect an external force and generate a signal indicative of the magnitude of such external force. The controller **34** is configured to command the airbag **35** to deploy based on the signal from one or more sensors **40**, such as the airbag sensor. Accordingly, the controller **34** is configured to determine when the airbag **35** has been deployed.

(13) The controller **34** includes at least one processor **44** and a non-transitory computer readable storage device or media **46**. The processor **44** may be a custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the controller **34**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a macroprocessor, a combination thereof, or generally a device for executing instructions. The computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using a number of memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or

combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **34** in controlling the vehicle **10**. The controller **34** of the vehicle **10** may be referred to as a vehicle controller and may be programmed to execute a method **100** (FIG. **2**) as described in detail below.

(14) The instructions may include one or more separate programs, each of which comprises an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the processor **44**, receive and process signals from the sensor system **28**, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the vehicle **10**, and generate control signals to the actuator system **30** to automatically control the components of the vehicle **10** based on the logic, calculations, methods, and/or algorithms. Although a single controller **34** is shown in FIG. **1**, embodiments of the vehicle **10** may include a plurality of controllers **34** that communicate over a suitable communication medium or a combination of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the vehicle **10**. In various embodiments, one or more instructions of the controller **34** are embodied in the control system **98**.

(15) The vehicle **10** includes a user interface **23**, which may be a touchscreen in the dashboard. The user interface **23** may include, but is not limited to, an alarm, such as one or more speakers **27** to provide an audible sound, haptic feedback in a vehicle seat or other object, one or more displays **29**, one or more microphones **31** and/or other devices suitable to provide a notification to the vehicle user of the vehicle **10**. The user interface **23** is in electronic communication with the controller **34** and is configured to receive inputs by a vehicle occupant **11** (e.g., a vehicle driver or a vehicle passenger). For example, the user interface **23** may include a touch screen and/or buttons configured to receive inputs from a vehicle occupant **11**. Accordingly, the controller **34** is configured to receive inputs from the user via the user interface **23**. The vehicle **10** may include one or more displays **29** configured to display information to the vehicle occupant **11** (e.g., vehicle operator or passenger) and may be a head-up display (HUD).

(16) The communication system **36** is in communication with the controller **34** and is configured to wirelessly communicate information to and from other remote vehicles **48**, such as but not limited to, other vehicles (“V2V” communication), infrastructure (“V2I” communication), remote systems at a remote call center (e.g., ON-STAR by GENERAL MOTORS) and/or personal electronic devices, such as a mobile phone. In the present disclosure, the term “remote vehicle” means a vehicle, such as a car, configured to transmit one or more signals to the vehicle **10** while not physically connected to the vehicle **10**. In certain embodiments, the communication system **36** is a wireless communication system configured to communicate via a wireless local area network (WLAN) using IEEE 802.11 standards or by using cellular data communication. However, additional, or alternate communication methods, such as a dedicated short-range communications (DSRC) channel, are also considered within the scope of the present disclosure. DSRC channels refer to one-way or two-way short-range to medium-range wireless communication channels specifically designed for automotive use and a corresponding set of protocols and standards. Accordingly, the communication system **36** may include one or more antennas and/or communication transceivers **37** for receiving and/or transmitting signals, such as cooperative sensing messages (CSMs). The communication transceivers **37** may be considered sensors **40**. The communication system **36** is configured to wirelessly communicate information between the vehicle **10** and another vehicle. Further, the communication system **36** is configured to wirelessly communicate information between the vehicle **10** and infrastructure or other vehicles.

(17) Referring to FIGS. **1** and **3**, the system **98** is configured to avoid impacts with an object **200** along a path **202** thereof by taking into consideration the friction between the tires **18** and the road **39**. When an impact is imminent with an object **200**, an impact may be avoided by automated emergency braking or automated evasive steering. At high speeds, avoiding an impact by steering

gives the driver or automated system more time margin to react, but this maneuver can cause other challenges if the road surface friction is low. The system **98** estimates surface friction when an impact is imminent and then uses that estimate to take the appropriate action to avoid an impact given the road conditions. Surface friction may be estimated by applying a short brake pulse when an impact is imminent, just before the brake-versus-steer decision needs to be made. The longitudinal response of the vehicle is used to estimate friction. If friction is low, the brake pulse is sustained and the system **98** transitions to automated emergency braking. If friction is adequate, automated evasive steering may engage, and the system **98** may use the friction estimate as a control parameter.

(18) With reference to FIGS. **2** and **3**, a method **100** is configured for friction-aware impact avoidance with another object **200** along a path **202** of the vehicle **10**. As non-limiting examples, the object **200** may be another vehicle, a stationary building, a pedestrian, etc. The method **100** begins at block **102**. At block **102**, the controller **34** receives sensor data from the sensors **40** (e.g., camera **41**). The sensor data may be indicative, among other things, of the location and movement of the objects around vehicle **10**. At block **102**, the vehicle controller **34** detects an object **200** that is along the path **202** of the vehicle **10** and that is within a predetermined distance from the vehicle **10** using the sensor data. Alternatively, or additionally, the vehicle **10** may use V2V or V2I communications to detect the object **200** that is along the path **202** of the vehicle **10** and that within a predetermined distance from the vehicle **10**. Then, the method **100** continues to block **104**.

(19) At block **104**, the vehicle controller **34** determines the time to impact (TTI) with the object **200** using, among other, vehicle speed and the distance from the vehicle **10** to the object **200**. The vehicle speed of the vehicle **10** and the distance from the vehicle **10** to the object **200** may be collected by the sensors **40**. The vehicle controller **34** also determines the last time to steer (LTTS). The LTTS is the shortest possible time before an impact that the vehicle **10** could steer and avoid an impact with the object **200**. The controller **34** may also determine the time to determine friction (TTDF), which may be a predetermined amount. The TTDF is the time it takes to determine the coefficient of friction between the road **39** and the tires **17**. As a non-limiting example, the TTDF may be between 300 and 500 microseconds. Then, at block **104**, the controller **34** determines the sum of the LTTS plus the TTDF. Next, the controller **34** compares the TTI with the sum of the LTTS and the TTDF. If the TTI is equal to or less than the sum of the LTTS and the TTDF, then the method **100** proceeds to block **106**. At block **106**, the controller **34** commands the vehicle **10** to autonomously fully brake immediately. In other words, at block **106**, the controller **34** commands the brake system **26** to immediately brake the vehicle **10**. If the TTI is greater than the sum of the LTTS and the TTDF, then the method **100** proceeds to block **108**.

(20) At block **108**, the controller **34** determines whether the last time to brake (LTTB) to avoid the object **200** along the path of the vehicle **10**. The LTTB is the shortest possible time before an impact that the vehicle **10** could brake and avoid an impact with the object **200**. The LTTB may be determined using, among other things, the sensor data from the sensors **40**. For example, the LTTB may be determined using the vehicle speed of the vehicle **10**, the distance from the vehicle **10** to the object **200**, and/or the speed of the object **200**. The controller **34** then determines the sum of the LTTB plus the TTDF. Next, the controller **34** compares the TTI with the sum of the LTTB and the TTDF. If the TTI is less than the sum of the LTTB and the TTDF, then the method **100** proceeds to block **110**. If the TTI is equal to or greater than the sum of the LTTB and the TTDF, then the method **100** returns to block **102**. At block **110**, the vehicle controller **34** commands the brake system **26** to apply a brake load to the wheels **15** to determine the coefficient of friction between the road **39** and the tires **17**. Thus, the initiation of the brake pulse at block **110** depends on the estimated TTI. The brake pulse may only be initiated in a time window that is less than the LTTB plus a buffer (i.e., TTDF) and greater than the LTTS. Specifically, at block **110**, the vehicle controller **34** may command the brake system **26** to apply a brake pulse (e.g., friction o regeneration) for a predetermined amount of time (e.g., 300 to 500 microsecond) to one or more

wheels **15** for the purpose of road friction estimation. The brake pulse may serve as an alert to the driver that a potential impact is imminent. This braking may be accompanied by other alerts to the driver. For example, at this juncture, the controller **34** may command the user interface **23** to provide a visual and/or audible notification indicative that an impact is imminent. The display **29**, for example, may issue an alert to the driver (e.g., “keep braking” and/or “do not steer”).

Inadequate road friction may be used as a condition to inhibit prompts or “nudges” to suggest that the driver attempt to steer evasively (e.g., alarms, vibrations, screen messages, HUD alerts, etc.). Inadequate road friction may be used as a condition to inhibit evasive steering assist that engages after the driver begins to steer. In this case, the system **98** may continue to apply the brakes and use automated steering commands to stabilize the vehicle **10** instead of avoiding the object **200**.

(21) In one embodiment, evasive steering only engages after a driver begins to steer and assists the driver in avoiding an impact. In this embodiment, the friction estimate may be used to as a condition to inhibit prompts to suggest that the driver attempts to steer evasively. For example, if the estimated coefficient of friction is less than a predetermined coefficient threshold, then the controller **34** may command the vehicle **10** to inhibit evasive steering assist that engages after the driver begins to steer.

(22) To determine the coefficient of friction between the road **39** and the tires **17**, the controller **34** monitors a longitudinal response (e.g., the longitudinal deceleration) of the vehicle **10** using the sensor data from the sensors **40** at the same time as the braking load is applied to the wheels **15** of the vehicle **10**. Using this longitudinal response (e.g., longitudinal deceleration), the controller **34** determines (i.e., estimates), in real-time, the coefficient of friction between the road **39** and the tires **17**. Then, the method **100** continues to block **112**. This friction estimation is used to decide the optimal impact avoidance control action (e.g., braking and/or steering). The control action may be a fully autonomous action or an assistant action by the vehicle **10**. After executing block **110**, the method **100** continues to block **112**.

(23) At block **112**, the controller **34** compares the coefficient of friction between the road **39** and the tire **17** of the vehicle **10** with a predetermined friction threshold. If the coefficient of friction between the road **39** and the tires **17** of the vehicle **10** is less than the predetermined friction threshold, then the method **100** continues to block **114**. At block **113**, the controller **34** commands the braking system **26** to continue applying a braking load to the wheels **15** to avoid or mitigate an impact with the object **200**. In other words, if the road friction is not sufficient for an evasive steering maneuver, the controller **34** commands the braking system **26** to continue applying a braking load to the wheels **15** to avoid or mitigate an impact with the object **200**. The magnitude of the braking could be different than the one used for detecting the coefficient of friction. The term “braking load includes brake load and/or a regen torque. If the coefficient of friction between the road **39** and the tires **17** of the vehicle **10** is equal to or greater than the predetermined friction threshold, then the method **100** proceeds to block **116**.

(24) At block **116**, the controller **34** determines an updated last time to steer (ULLTS) to avoid the object **200** that is along the path **202** of the vehicle **10** based on the coefficient of friction between the road **39** and the tire **17** of the vehicle **10**. In other words, the controller **34** recalculates the LTTS. Then, the method **100** continues to block **118**.

(25) At block **118**, the controller **34** compares the TTI with the ULLTS. If the TTI is equal to or less than the ULLTS, then the method **100** proceeds to block **120**. If the TTI is greater than the ULTS, then the method **100** returns to block **112**.

(26) At block **120**, the controller **34** commands the steering system **24** to initiate an automated steering maneuver to avoid the object **200** along the path **202**. A classification of the road friction (e.g., medium versus high or wet versus dry asphalt) may be used to modify the timing or shape of the evasive steering maneuver. When road friction is lower, the evasive steering maneuver may be initiated earlier to accommodate a lower curvature path. Thus, at block **120**, the controller **34** may determine a steering path and the radius of curvature of the steering path. The radius of curvature of

the steering path may be based on the coefficient of friction between the tire **17** and the road **39**. A continuous estimate of the road friction may be used in path planning or control. For instance, the estimated friction, or the maximum lateral and longitudinal accelerations derived therefrom, may be applied as constraints in path-planning to shape the desired trajectory according to the tire limits. Further, the estimated friction coefficient may be used as a model parameter in model-predictive control or other model/parameter-based control methodology, and/or as a variable used to compute the gains for any feedback controller. Also, the maximum lateral acceleration derived from the estimated friction coefficient may be applied as a constraint in control.

(27) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the presently disclosed system and method that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and can be desirable for particular applications.

(28) The drawings are in simplified form and are not to precise scale. For purposes of convenience and clarity only, directional terms such as top, bottom, left, right, up, over, above, below, beneath, rear, and front, may be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the disclosure in any manner.

(29) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to display details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the presently disclosed system and method. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(30) Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by a number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with a number of systems, and that the systems described herein are merely exemplary embodiments of the present disclosure.

(31) For the sake of brevity, techniques related to signal processing, data fusion, signaling, control,

and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

(32) This description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims.

Claims

1. A method controlling a vehicle, comprising: detecting that an object is along a path of the vehicle and within a predetermined distance from the vehicle; determining a time to impact with the object that is along the path of the vehicle and within the predetermined distance from the vehicle; determining a time to determine friction; determining a last time to steer the vehicle to avoid the object along the path of the vehicle; determining a last time to brake to avoid the object along the path of the vehicle; determining a sum of the time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle; determining a sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction; comparing the time to impact with the object that is along the path of the vehicle with the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction to determine whether the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction; in response to detecting the object that is along the path of the vehicle and within the predetermined distance from the vehicle and determining that the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction, determining, in real time, a coefficient of friction between a road and a tire of the vehicle, wherein determining, in real time, a coefficient of friction between a road and a tire of the vehicle comprises: applying a braking load to a wheel of the vehicle; monitoring a movement of the vehicle at the same time as the braking load is applied to the wheel of the vehicle; and determining, in real time, the coefficient of friction between the road and the tire of the vehicle based on the movement of the vehicle while the braking load; comparing the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold; in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold, determining an updated last time to steer to avoid the object that is along the path of the vehicle based on the coefficient of friction between the road and the tire of the vehicle; comparing the time to impact with the object along the path of the vehicle with the updated last time to steer to avoid the object that is along the path of the vehicle to determine whether the time to impact with the object along the path of the vehicle is greater than the updated last time to steer to avoid the object that is along the path of the vehicle; and in response to determining the coefficient of friction between the road and the tire of the vehicle, commanding the vehicle to perform a control action, wherein the control action is steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold.

2. The method of claim 1, further comprising: comparing the coefficient of friction between the

road and the tire of the vehicle with a predetermined friction threshold to determine whether the comparing the coefficient of friction between the road and the tire of the vehicle is less than the predetermined friction threshold; and wherein the control action is continuously applying the braking load to the wheel of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is less than the predetermined friction threshold.

3. The method of claim 1, further comprising: comparing the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold; and wherein the control action is steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold.

4. The method of claim 3, further comprising: determining a steering path to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold, wherein the steering path has a radius of curvature; and wherein the determining the steering path to avoid the object along the path of the vehicle includes determining a radius of curvature of the steering path; and wherein the radius of curvature of the steering path is based on the coefficient of friction between the tire and the road.

5. The method of claim 1, further comprising: in response to determining that the time to impact with the object that is along the path of the vehicle is greater than the sum of the time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle, determining the last time to brake to avoid the object along the path of the vehicle; in response to determining that the time to impact with the object that is along the path of the vehicle is less than the sum of the time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle, determining a sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction; and in response to determining that the time to impact with the object that is along the path of the vehicle is less than the sum of the last time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle, comparing the time to impact with the object that is along the path of the vehicle with the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction to determine whether the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction.

6. The method of claim 1, wherein inadequate road friction is used as a condition to inhibit evasive steering assist that engages after a vehicle driver begins to steer.

7. A system for controlling a vehicle, comprising: a plurality of sensors; a controller in communication with the plurality of sensors, wherein the controller is programmed to: detect that an object is along a path of the vehicle and within a predetermined distance from a vehicle; determine a time to impact with the object that is along the path of the vehicle and within the predetermined distance from the vehicle; determine a time to determine friction; determine a last time to steer the vehicle to avoid the object along the path of the vehicle; determine a last time to brake to avoid the object along the path of the vehicle; determine a sum of the time to determine friction and the last time to steer the vehicle to avoid the object along the path of the vehicle; determine a sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction; compare the time to impact with the object that is along the path of the vehicle with the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction to determine whether the time to impact is less than the sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction; in response to detecting the object that is along the path of the vehicle and within the predetermined distance from the vehicle and determining that the time to impact is less than the

sum of the last time to brake to avoid the object along the path of the vehicle and the time to determine friction, determine, in real time, a coefficient of friction between a road and a tire of the vehicle, wherein to determine, in real time, a coefficient of friction between a road and a tire of the vehicle, the controller is programmed to: command a braking system to apply a braking load to a wheel of the vehicle; monitor a movement of the vehicle at the same time as the braking load is applied to the wheel of the vehicle; and determine, in real time, the coefficient of friction between the road and the tire of the vehicle based on the movement of the vehicle while the braking load; compare the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold; in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold, determine an updated last time to steer to avoid the object that is along the path of the vehicle based on the coefficient of friction between the road and the tire of the vehicle; compare the time to impact with the object along the path of the vehicle with the updated last time to steer to avoid the object that is along the path of the vehicle to determine whether the time to impact with the object along the path of the vehicle is greater than the updated last time to steer to avoid the object that is along the path of the vehicle; and in response to determining the coefficient of friction between the road and the tire of the vehicle, command the vehicle to perform a control action, wherein the control action is steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold.

8. The system of claim 7, wherein the controller is programmed to: compare the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is less than the predetermined friction threshold; and wherein the control action is continuously applying the braking load to the wheel of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is less than the predetermined friction threshold.

9. The system of claim 7, wherein the controller is programmed to: compare the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold; and wherein the control action is steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold.

10. The system of claim 7, wherein the controller is programmed to: compare the coefficient of friction between the road and the tire of the vehicle with a predetermined friction threshold to determine whether the coefficient of friction between the road and the tire of the vehicle is or equal to or greater than the predetermined friction threshold; and wherein the control action is steering the vehicle to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold.

11. The system of claim 10, wherein the controller is programmed to: determine a steering path to avoid the object along the path of the vehicle in response to determining that the coefficient of friction between the road and the tire of the vehicle is equal to or greater than the predetermined friction threshold, wherein the steering path has a radius of curvature; and determining a radius of curvature of the steering path; and wherein the radius of curvature of the steering path is based on the coefficient of friction between the tire and the road.

12. The system of claim 7, wherein the control action includes commanding a display of the vehicle to display an alert.

13. The system of claim 7, wherein inadequate road friction is used as a condition to inhibit prompts to suggest that a vehicle driver attempt to steer evasively.
